

P. Shubham Parashar, Ph.D.

Theoretical Condensed Matter Physics | Quantum Transport and Kinetic Theory

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Research Profile

Theoretical condensed-matter physicist developing analytical and computational descriptions of transport in spatially inhomogeneous two-dimensional electron systems. My work connects semiclassical Boltzmann kinetics, electron hydrodynamics, magnetotransport, quantum Hall response, and finite-resolution inverse problems. Recent projects address higher-harmonic corrections to hydrodynamic closure, observability in scanning-probe transport, and stochastic worldline mechanisms for heavy-tailed trajectory geometry.

Education and Research Experience

- 2022–2026 **Ph.D., Physics**, University of California San Diego. Advisor: Michael M. Fogler. Dissertation: *Electric and Magnetic Response near Perturbations in Two-Dimensional Electron Gases*.
- 2017–2021 Research with Benjamin Grinstein and Congjun Wu, UC San Diego, on non-associative operators and twist operators on S^7 backgrounds with G_2 instantons.
- 2016–2017 **M.S., Physics**, Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.
- 2012–2016 **B.S., Physics**, Indian Institute of Science, Bangalore. Advisor: Rahul Pandit.
- 2015–2016 Research Intern, Japan Advanced Institute of Science and Technology. Advisor: Ryo Maezono; MEXT-supported.

Publications and Manuscripts

Peer-reviewed

Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas

P. Shubham Parashar and Michael M. Fogler

Physical Review B 113, 155443 (2026). arXiv:2601.16402.

Nanoscale imaging of magnetotransport around a circular p–n junction in graphene

Z. J. Krebs, W. A. Behn, K. J. Smith, M. A. Fortman, K. Watanabe, T. Taniguchi, P. Shubham Parashar, M. M. Fogler, and V. W. Brar

Physical Review Letters 136, 136301 (2026); Editors' Suggestion. arXiv:2409.19468.

Under review / submitted

Fourth-order closure obstruction and chiral nonlocality in circular kinetic magnetotransport

P. Shubham Parashar

Under review at *Physical Review B* (2026). arXiv:2607.22730.

Thermal and viscous contrast in quantum Hall scanning-probe images

P. Shubham Parashar

Submitted to *Physica E: Low-dimensional Systems and Nanostructures* (2026). arXiv:2607.16315.

Worldline optimal fluctuations and diffusion-defect shifts of denominator-tail exponents

P. Shubham Parashar

Manuscript under review (2026).

Selected Contributions

- Derived magnetotransport around smooth density perturbations across Ohmic and hydrodynamic regimes, including Landauer resistivity dipoles and Gurzhi-scale redistribution.
- Identified the fourth-order current operator omitted by stress-level hydrodynamic closure and tracked its chiral finite-wave-number completion into the nonlocal regime.
- Formulated finite-resolution observability tests for thermoelectric and Hall-viscous response in quantum Hall scanning-probe measurements.

Awards, Methods, and Selected Presentations

Awards: Bhadra Fellowship (2024); NSF-supported graduate research (2018–2020); UC San Diego Scholar's Grant (2017); KVPY Fellowship (2012–2017); MEXT-supported research internship (2015–2016).

Methods: Boltzmann-harmonic hierarchies, hydrodynamic and higher-gradient closures, Green-function and boundary-value methods, asymptotic and Mellin analysis, stochastic path integrals, inverse problems, convergence and uncertainty tests.

Tools: Python, MATLAB, Mathematica/Wolfram Language, C++, Fortran, L^AT_EX, Git.

Selected talks: Ph.D. dissertation defense, UC San Diego (2026); APS March Meeting (2020, 2018); Columbia University (2022).